At position 1, where θ = 0 degrees, the plane of the coil is perpendicular to the direction of lines of the flux. Now, the flux linked with the coil is maximum, but the rate of change of the flux linkages is minimum. So, no emf is induced in the coil. i.e., is at the starting position the emf induced is zero.

When the coil is rotated further, the rate of change of flux linkage increases up to the position 3, where θ = 90 degrees. At this position, the plane of the coil is parallel to the lines of flux. Now, the flux linked with the coil is minimum, but rate of change of flux linkage is maximum. Therefore, at this position emf induced in the coil is maximum. On further rotation the value of generated emf continues to decrease from maximum to minimum value.

At position 5, where θ = 180 degree, the generated emf is reduced to zero. The magnitude of emf with respect to the coil position is represented in fig 7.1. from position 5 to position 7 (that is 180 degrees to 270 degrees), the induced emf value starts again from zero to maximum and from position 7 to positon 1 (from 270 degrees to 360 degrees) maximum to zero in opposite direction.

In this way the generated emf goes on one cycle as alternating emf. The two halved split-rings make the bidirectional emf into unidirectional emf to behave as DC generator. This set up is shown in fig 7.2. In the first half revolution current flows along the path A-B-a-M-L-b-C-D-A, (fig 7.2a). The coil in the second half rotation, the induced currents are reversed. But the positions of brushes 'a and b' are also reversed so that current through the load remains the same direction, that is the loop D-C-b-M-L-a-B-A-D (fig 7.2b). The device making the alternating supply into unidirectional supply is by means of a device called 'commutator'.

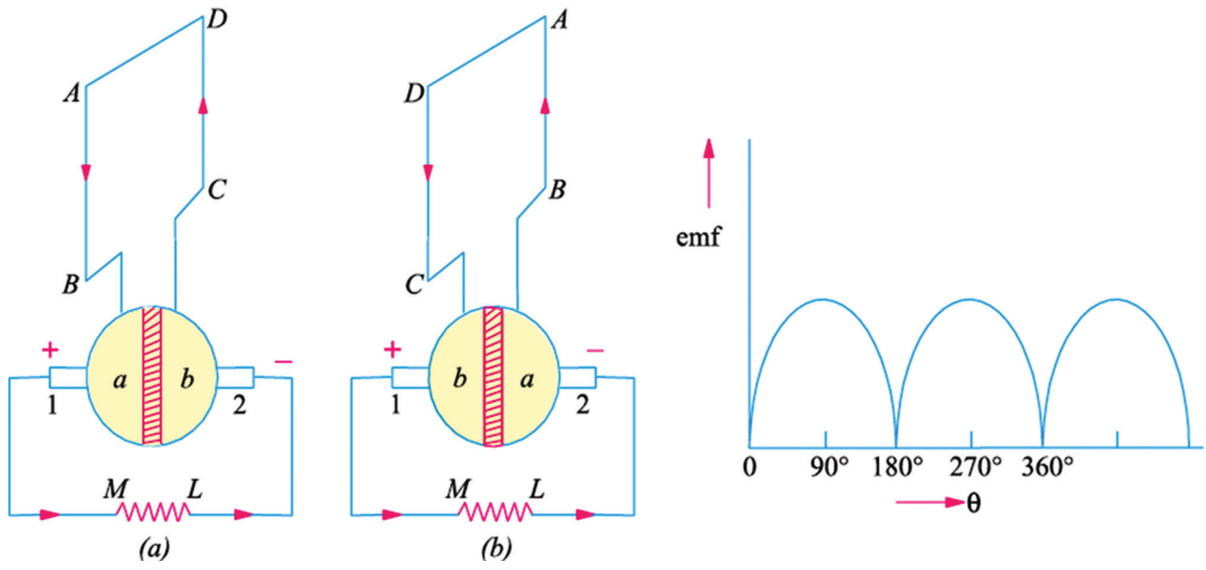


Fig 7.2 *AC into DC Waveform*

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